

Data Imbalance in Machine Learning

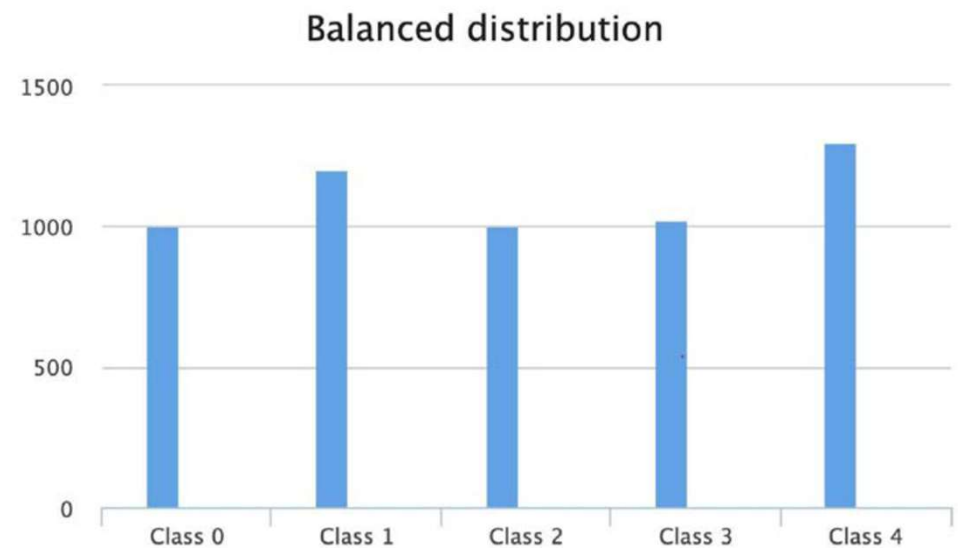
BY

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Balanced datasets

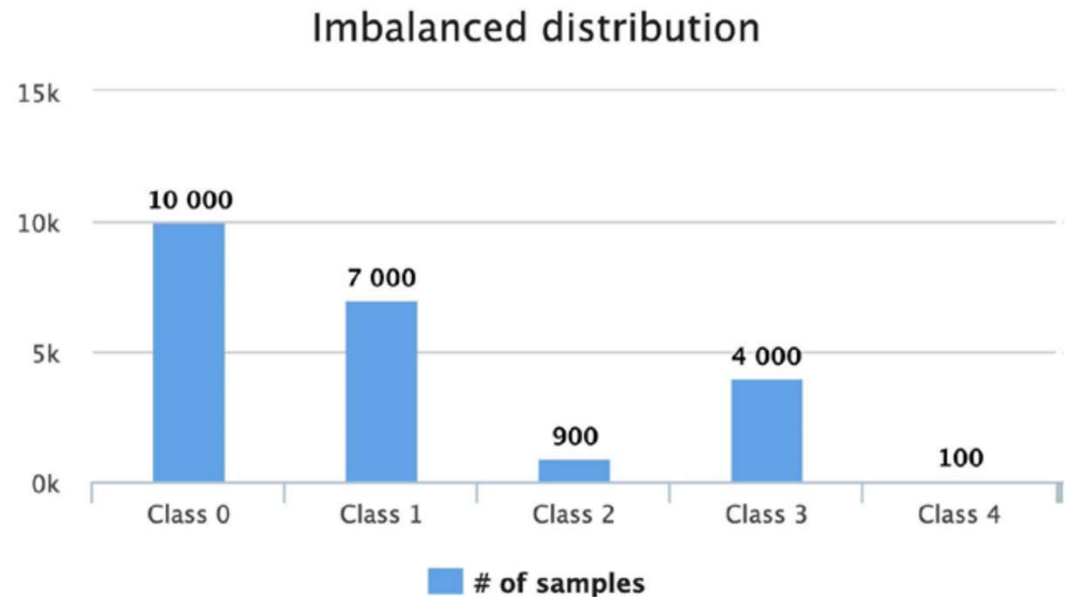
- A dataset can be said to have a balanced distribution when all the target classes have a similar number of examples.



Balanced distribution with an almost equal number of examples for each class

Imbalanced/Skewed datasets

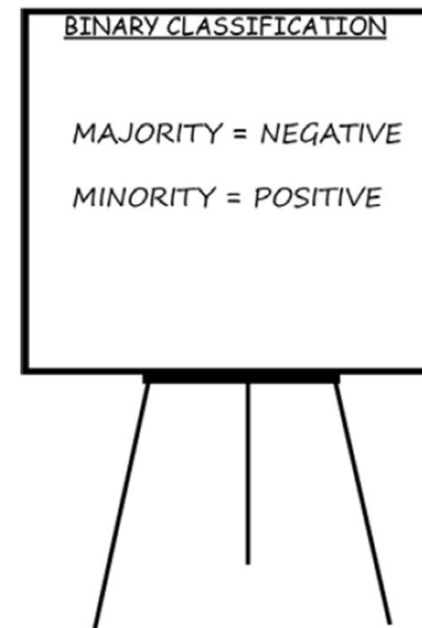
- Imbalanced datasets or skewed datasets are those that have some target classes (also called labels) that outnumber the rest of the classes



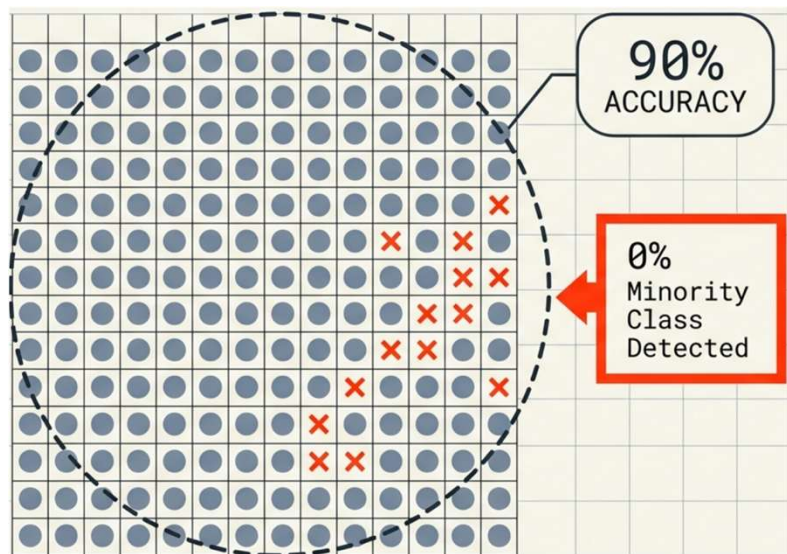
An imbalanced dataset with five classes and a varying number of samples

Majority vs Minority Class

- We label the class with more instances as the “majority” or “negative” class and the one with fewer instances as the “minority” or “positive” class.
- Most of the time, our main interest lies in the minority class, which is why we often refer to the minority class as the “positive” class and to the majority class as the “negative” class:

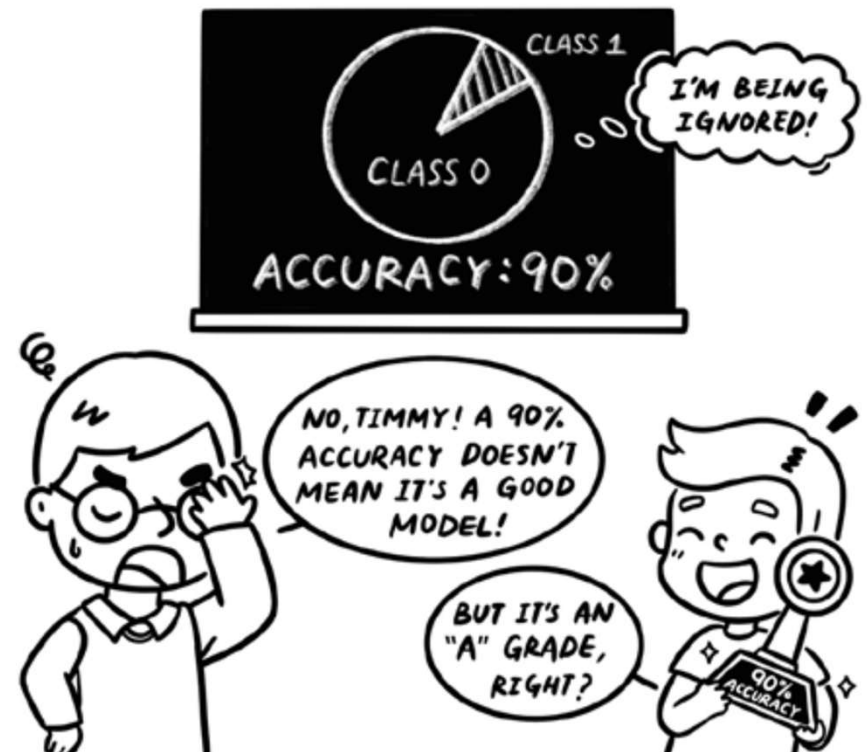


The Accuracy Trap Hides Critical Failures in Skewed Data



- A model trained on a 1:9 dataset can blindly predict the majority class for every instance and still achieve 90% accuracy.

WHEN ACCURACY ISN'T SO ACCURATE



Situations causes imbalance in datasets

- **Inherent in the problem**
- **High cost of data collection while bootstrapping a machine learning solution**
- **Noisy labels for certain classes**
- **Labeling errors**
- **Sampling bias**
- **Data cleaning**

Inherent in the problem

- Sometimes, the task we need to solve involves detecting outliers in datasets.
- For example, patients with a certain disease or fraud cases in a set of transactions.
- In such cases, the dataset is inherently imbalanced because the target events are rare to begin with.

High cost of data collection while bootstrapping a machine learning solution

- The cost of collecting data might be too high for certain classes.
 - For example, collecting data on COVID-19 patients incurs high costs due to the need for specialized medical tests, protective equipment, and the ethical and logistical challenges of obtaining informed consent in a high-stress healthcare environment.

Noisy labels for certain classes

- This may happen when a lot of noise is introduced into the labels of the dataset for certain classes during data collection

Labeling errors

- Errors in labeling can also contribute to data imbalance.
- For example, if some samples are mistakenly labeled as negative when they are positive, this can result in an imbalance in the dataset.
- Additionally, if a class is already inherently rare, human annotators might be biased and overlook the few examples of that rare class that do exist.

Sampling bias

- Data collection methods can sometimes introduce bias in the dataset.
- For example, if a survey is conducted in a specific geographical area or among a specific group of people, the resulting dataset may not be representative of the entire population.

Data cleaning

- During the data cleaning or filtering process, some classes or samples may be removed due to incomplete or missing data.
- This can result in an imbalance in the remaining dataset.

Why can imbalanced data be a challenge?

- What is Imbalanced Data?
- Misleading Evaluation Metrics.
- Bias in Loss Function.
- Unequal Misclassification Costs.
- Computational Constraints.
- Lack of Diversity in Minority Class.
- Poorly Calibrated Models.
- Fixed Threshold Problem.

1. What is Imbalanced Data?

- A dataset where one class significantly outnumbers the other(s)
- Example:
 - Fraud detection: 99% normal, 1% fraud
 - Medical diagnosis: rare disease cases

2. Misleading Evaluation Metrics

- Problem: Accuracy becomes unreliable
 - Example:
 - 99% non-fraud → model predicts all “non-fraud”
 - Accuracy = 99% (but useless model!)
- Root Cause:
 - Dominance of True Negatives (TN) in the confusion matrix
- Better Alternatives:
- Precision, Recall, F1-score, PR Curve / ROC Curve
- These metrics depend on a classification threshold, which must be optimized, not assumed (e.g., not always 0.5)

3. Bias in Loss Function

- Problem: Standard loss functions aim to minimize overall error
- Majority class dominates → model ignores minority class
- Result:
- High accuracy but poor minority detection
- Key Insight: Model becomes biased toward majority class

4. Unequal Misclassification Costs

- Problem: Errors are not equally costly
- Example:
- Cancer diagnosis:
 - False Negative (miss cancer) → very costly
 - False Positive → less costly (extra tests)
- Insight: Need to consider cost-sensitive learning

5. Computational Constraints

- **Problem: Large datasets = high training cost**
- **Imbalance → majority class dominates storage & processing**
- **Practical Issues:**
 - Memory limitations
 - Long training time
- **Common Solutions:**
 - Undersampling (reduce majority class)
 - Selective data handling

6. Lack of Diversity in Minority Class

- Problem: Minority class may have:
 - Few samples
 - Low variation
- Result: Model cannot learn proper decision boundary
- Example:
- Computer vision:
 - Few images of rare objects → poor detection
- Solution:
 - Data Augmentation

7. Poorly Calibrated Models

- What is Calibration?
 - Predicted probability should reflect true likelihood
- Problem:
 - Imbalanced data → biased probability estimates
- Effects:
 - Overconfident wrong predictions
 - Poor minority detection
- Solutions:
 - Platt Scaling
 - Isotonic Regression

8. Fixed Threshold Problem

- Problem:
- Default rule:
 - Predict class = 1 if probability > 0.5
 - Not optimal for imbalanced data!
- Better Approach:
 - Use PR Curve to choose threshold
- Insight:
 - Threshold = hidden hyperparameter

When Should We *Not* Worry About Imbalanced Data?

- When Performance Impact is Minimal
- When the Goal is to Predict the Majority Class
- When Misclassification Costs Are Similar
- When Dataset is Large Enough

1. When Performance Impact is Minimal

- Scenario: Imbalance exists, but model performance is still acceptable
- Insight:
- Not all imbalances require intervention
- Sometimes, baseline models already perform well
- Key Message:
 - Avoid over-engineering if imbalance does not significantly harm performance

2. When the Goal is to Predict the Majority Class

- Scenario: Business objective focuses on majority class prediction
- Example:
- Online advertising:
 - Focus = users likely to click (majority)
 - Less concern = users who dislike ads (minority)
- Insight:
 - Minority class may have low business value

3. When Misclassification Costs Are Similar

- Scenario: False positives $\approx$ False negatives (similar impact)
- Example:
- Email spam detection:
 - Missing a spam email occasionally is acceptable
 - Misclassifying is not critical
- Insight:
 - No strong need for cost-sensitive adjustments

4. When Dataset is Large Enough

- Scenario: Even with imbalance (e.g., 1:100), dataset contains:
 - Large number of minority samples
 - Rich representation of patterns
- Benefit:
 - Model can still learn minority class effectively
- Important Note: Always compare:
 - Baseline model
 - Improved models (e.g., sampling, threshold tuning)

5. Recommended Best Practice

- Even when imbalance seems harmless:
 - Compare multiple approaches:
 - Baseline model
 - Threshold tuning
 - Oversampling
 - Undersampling
- Cost-sensitive learning
- Key Insight: “Do not assume imbalance is harmless—validate it empirically.”

Introduction to the imbalanced-learn library

```
pip3 install imbalanced-learn==0.11.0
```

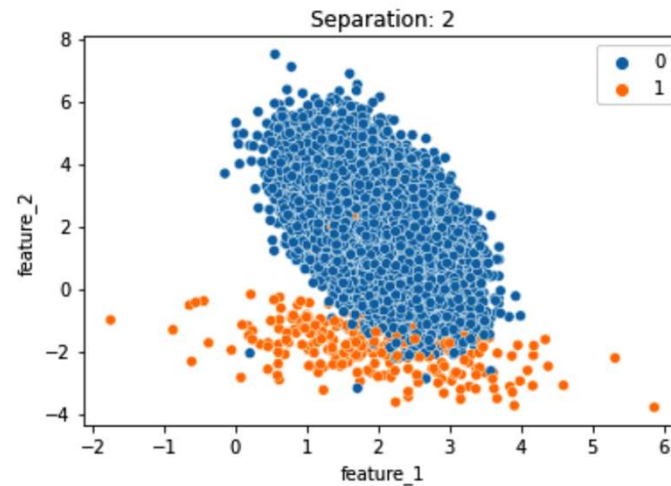
```
from sklearn.datasets import make_classification
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def make_data(sep):
    X, y = make_classification(n_samples=50000,
                              n_features=2, n_redundant=0,
                              n_clusters_per_class=1, weights=[0.995],
                              class_sep=sep, random_state=1)

    X = pd.DataFrame(X, columns=['feature_1', 'feature_2'])
    y = pd.Series(y)
    return X, y
```

```
from collections import Counter

X, y = make_data(sep=2)
print(y.value_counts())
sns.scatterplot(data=X, x="feature_1", y="feature_2", hue=y)
plt.title('Separation: {}'.format(separation))
plt.show()
```



0	49498
1	502

Conclusion

- “Imbalanced data is not always a problem, it becomes a problem only when it affects decision quality.”